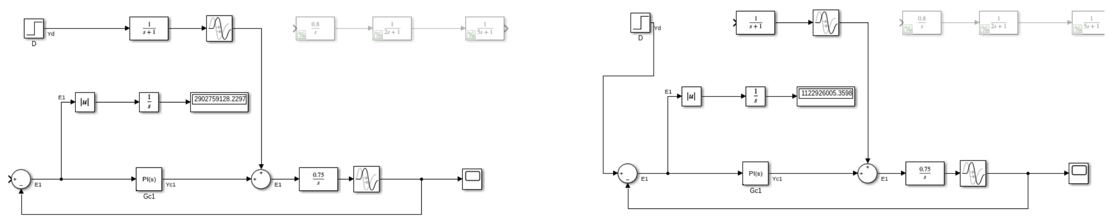
CHE 565 -- Exam 2

Soki Sem

May 5, 2026

Problem 1

A



(a) Disturbance Diagram (b) Setpoint Diagram

Figure 1: Block Diagram Control System

$$\text{Process Transfer Function:} \frac{0.8}{s\,(2s+1)\,(5s+1)}$$

$$\text{Modeled Process Transfer Function:} \frac{0.75e^{-4s}}{s}$$

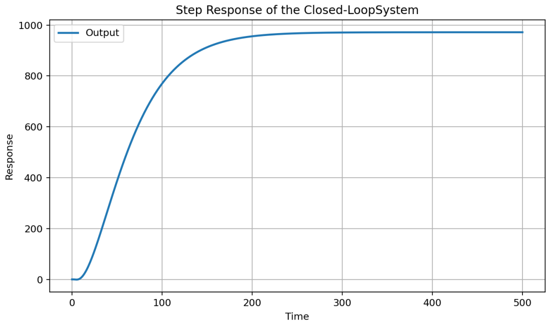
$$\text{Disturbance Transfer Function:} \frac{e^{-s}}{s+1}$$

Using lambda tuning for PI control, we find the controller parameters as follows:

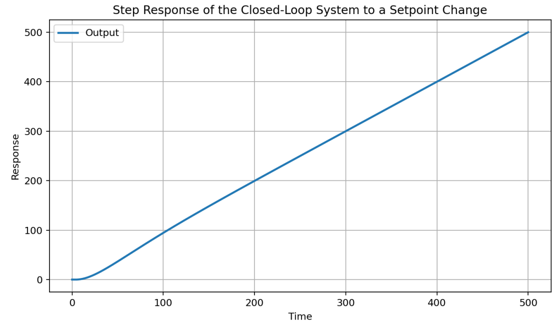
$$K' = \frac{K_p}{\tau_p}, Kc = \frac{\tau_p}{K'(\lambda + \theta_d)^2}$$
$$\tau_I = \tau_p$$
$$\lambda = 8\theta_d$$
$$\tau_D = 0$$

Where τ_p= 1 θ_d= 4 K_p= 0.75

K_c := 0.0700 τ_I := 68.0000 λ := 32.0000 τ_D := 0.0000 And an IAE of: 417965.4725 for the setpoint change and 123704.5080 for the disturbance.



(a) Disturbance



(b) Setpoint Change

Figure 2: Closed-Loop Step Responses